Explicit Order–Degree Curves for Differential D-Finite Telescoping

A connection-matrix refinement addressing Chen–Kauers Problem 3

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Abstract

For a bivariate differential D-finite input presented by rank-m rational connection matrices A/D and B/D, we give an explicit sufficient order–degree curve using only the maximum bidegrees (t, s) of A, B, D. For every integer $r \geq m(s+1)$, a telescoper of exact order r exists with degree at most

$$\left\lfloor \frac{mt[(2r-1)s+r]}{r+1-m(s+1)} \right\rfloor .$$

A polynomial certificate support bound and an integer coefficient bound are obtained simultaneously. The argument counts polynomial unknowns in the generalized Apagodu–Zeilberger ansatz and controls the certificate-only kernel by evaluation at an ordinary point. It imposes no Fuchsian or regular-singularity assumption. The result addresses the differential setting of Chen–Kauers Problem 3; it does not establish a first resolution, a new foundational principle, or an extension to arbitrary discrete and mixed Ore systems. Seven exact ansatz computations and three special-function certificate identities accompany the proof. Their verification scripts are embedded in the single TeX source.

1 Question and contribution

Chen and Kauers' original arXiv Problem 3 asks for an order–degree curve for general D-finite functions [1]. Their definition permits a sufficient bound above an order threshold; it does not demand the true minimum order or a sharp boundary. This note supplies a concrete formula in the full bivariate *differential* setting. The problem's surrounding discussion also includes summation, which is not covered here.

The finite input used in the theorem consists of derivative multiplication matrices. It is available from any finite-rank annihilating ideal by standard normal forms. No telescoper or reduced integration quotient must be computed to evaluate the bound. This separates the result from a bound that uses the output operator itself as an input parameter.

Attribution and status. The vector ansatz and certificate-kernel subtraction already occur in Chen–Kauers–Koutschan [2, Theorem 6]; the present calculation tracks both polynomial degrees explicitly. Kauers' book discusses order–degree bounds in its broader treatment of creative telescoping [3, Section 5.5]. Van der Hoeven's reduction framework gives polynomial complexity and degree control for nonminimal differential telescopers in its representation [4, Section 9]. Consequently, general degree control is not itself a new discovery here. The particular direct formula below was not identified in the sources examined, but that observation does not establish historical priority. This manuscript presents a proved quantitative refinement, not a certified first solution of a currently open problem.

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What is proved. The proof covers every finite-rank differential presentation over a characteristic-zero field, including irregular singularities. It also provides explicit support bounds for a certificate and, over the rationals, an arithmetic extension of the rank-one construction. The proof and finite checks have been reviewed during preparation; they have not undergone external peer review or proof-assistant formalization.

2 An input-level order–degree curve for differential D-finite functions

Chen and Kauers' original Problem 3 asks for an order–degree curve for general D-finite functions. We establish such a curve for every bivariate *differential* D-finite input, expressed in its rational connection matrices. These matrices are computed from the input annihilating ideal; no telescoper or reduced telescoping quotient is assumed available. The construction refines the generalized Apagodu–Zeilberger polynomial ansatz of Chen, Kauers, and Koutschan [2] by tracking both polynomial degrees. It is not asserted to be a historically new methodological principle. Discrete and mixed Ore variants are outside the theorem.

2.1 Input representation

Let $\mathbb{K}$ be a characteristic-zero field, and put $\mathcal{R} = \mathbb{K}(x,y)\langle\partial_x, \partial_y\rangle$. Suppose first that a nonzero D-finite function f is given together with a basis $F = (F_1, \dots, F_m)^T$ of its derivative module over $\mathbb{K}(x,y)$, with $F_1 = f$. Such a basis exists: the derivative module is finite-dimensional and contains the nonzero element f. Every F_j is an operator in $\mathcal{R}$ applied to f, so rational row combinations of F are permitted certificate functions. Clearing denominators gives

$$\partial_x F = \frac{A}{D}F, \qquad \partial_y F = \frac{B}{D}F, \qquad A, B \in \mathbb{K}[x,y]^{m\times m}, \quad 0 \neq D \in \mathbb{K}[x,y]. \tag{1}$$

The same representation is effective from a proper D-finite left ideal $I \subseteq \mathcal{R}$ of finite rank m. Compute a left Gröbner basis and enumerate its standard differential monomials $b_1 = 1, b_2, \dots, b_m$. Normal forms of $\partial_x b_j$ and $\partial_y b_j$ give the two multiplication matrices. A common denominator supplies (1) in the formal quotient module $\mathcal{R}/I$, with $F_j = [b_j]$ and $F_1 = [1]$. If I is merely contained in the annihilator of a particular function, the evaluated $b_j(f)$ may be dependent; this causes no problem, because the formal identity modulo I implies the identity on f. If $I = \mathcal{R}$, the represented function is zero and $P = 1$ is trivial.

The parameters below are explicit in this computed connection presentation. We do not claim a conversion bound solely in terms of the degrees of an arbitrary unprocessed PDE generating set. All presentation steps precede the telescoping calculation and use only the supplied D-finite ideal. Over an effectively computable coefficient field, they are ordinary finite normal-form and rational-function computations.

2.2 The order–degree theorem

Theorem 2.1. *Assume (1), with $m \geq 1$, and let $t, s \geq 0$ be integer upper bounds on the x- and y-degrees of D and every entry of A, B. Let r, d be nonnegative integers such that*

$$r + 1 > m(s+1), \qquad \big[r + 1 - m(s+1)\big](d+1) > mt\big[(2r-1)s + r\big]. \tag{2}$$

Set

$$a = d + (r-1)t, \qquad b = (r-1)s. \tag{3}$$

There exist polynomials $p_0, \dots, p_r \in \mathbb{K}[x]$, not all zero, and a row $U \in \mathbb{K}[x,y]^{1\times m}$ such that

$$\deg_x p_i \le d, \qquad \deg_x U_j \le a, \qquad \deg_y U_j \le b, \tag{4}$$

$$\sum_{i=0}^{r} p_i(x)\partial_x^i F_1 = \partial_y\left(D^{1-r}UF\right). \tag{5}$$

In the formal quotient presentation, this states $P - \partial_y Q \in I$, where

$$P = \sum_{i=0}^{r} p_i(x)\partial_x^i \neq 0, \qquad Q = D^{1-r}\sum_{j=1}^{m} U_j b_j \in \mathcal{R}.$$

Consequently, for every integer $r \ge m(s+1)$ there is a telescoper of exact *order r and degree at most $\lfloor c(r)\rfloor$, where*

$$\boxed{c(r) = \frac{mt\big[(2r-1)s+r\big]}{r+1-m(s+1)}.} \tag{6}$$

Proof. Let $v_0 = (1,0,\dots,0)$ and recursively define polynomial rows

$$v_{i+1} = D\partial_x v_i + v_i A - i(\partial_x D)v_i. \tag{7}$$

By induction, $\partial_x^i F_1 = D^{-i}v_i F$ and every entry of v_i has bidegree at most (it, is). Put $w_i = v_i D^{r-i}$, whose entries have bidegree at most (rt, rs).

Make an ansatz with the degree bounds (4). The coordinate identity sufficient for (5) is

$$\sum_{i=0}^{r} p_i w_i = D\partial_y U + UB - (r-1)(\partial_y D)U. \tag{8}$$

Every coordinate has bidegree at most $(d+rt, rs)$. With

$$n_P = (r+1)(d+1), \qquad n_U = (a+1)(b+1), \qquad E = (d+rt+1)(rs+1),$$

coefficient comparison yields a homogeneous linear system over $\mathbb{K}$ with at most mE equations and $n_P + mn_U$ unknowns.

We exclude solutions supported only on the certificate coordinates. Such a row U satisfies

$$\partial_y U = U\frac{(r-1)(\partial_y D)I_m - B}{D}. \tag{9}$$

Choose $y_0 \in \{0,1,\dots,s\}$ with $D(x,y_0) \neq 0$ in $\mathbb{K}[x]$; at least one exists because $\deg_y D \le s$. Over $\mathbb{K}(x)$ the coefficient matrix in (9) is regular at y_0. Formal power-series substitution around y_0 shows uniqueness for a prescribed value $U(x,y_0)$: the coefficient of $(y-y_0)^{j+1}$ is determined by earlier coefficients after division by the nonzero integer $j+1$. Therefore evaluation at y_0 injects the bounded polynomial solution space into $(\mathbb{K}[x]_{\le a})^m$. Its dimension over $\mathbb{K}$ is at most $m(a+1)$.

The full solution space has dimension at least $n_P + mn_U - mE$, and direct expansion gives

$$\begin{aligned} n_P + mn_U - mE - m(a+1) = {}& \big[r+1-m(s+1)\big](d+1) \\ & - mt\big[(2r-1)s+r\big] > 0. \end{aligned}$$

It consequently contains a vector with some $p_i \neq 0$, proving the first assertion. Notice that only coordinate matrices enter the argument; no analytic regularity of the solution functions is required.

For the curve, choose $d = \lfloor c(r)\rfloor$. Then $d+1 > c(r)$, so (2) holds. The constructed operator may have order $k < r$. In that case replace P by $\partial_x^{r-k}P$ and Q by $\partial_x^{r-k}Q$. Commutation of the two derivations preserves the certificate identity. The new operator has order exactly r, and its coefficient degrees do not increase, since left multiplication by ∂_x only differentiates existing coefficients or shifts their operator indices. This proves (6) with the exact-order convention. □

Proposition 2.2. *Two uniform admissible pairs in Theorem 2.1 are*

$$r = m(s+1), \qquad d = mt\big[m(s+1)(2s+1) - s\big],$$
$$r = 2m(s+1) - 1, \qquad d = t(4ms + 2m - 1).$$

The second gives order $O(m(s+1))$ and degree $O(mt(s+1))$. The curve (6) has horizontal asymptote $mt(2s+1)$.

Proof. For the first pair the surplus in the strict inequality of (2) is one. For the second it is $m(s+1+2ts) > 0$. The asymptote follows by dividing the numerator and denominator of (6) by r. □

Remark 2.3. The formula gives a sufficient region and does not identify the true minimum order, optimal degree, or a generic sharp curve. Neither $t = 0$ nor $s = 0$ needs exclusion. The exact-order adjustment in the proof preserves the degree bound, but it is not asserted to preserve the original certificate box in (4).

2.3 An arithmetic extension over the rationals

Proposition 2.4. *Suppose A, B, D have integer coefficients. Define*

$$\alpha = \max_i \sum_j \|A_{ij}\|_1, \qquad \beta = \max_i \sum_j \|B_{ij}\|_1, \qquad \delta = \|D\|_1,$$
$$H_P = \max_{0\le i\le r} \left\{ \delta^{r-i} \prod_{j=0}^{i-1} (\alpha + 2jt\delta) \right\},$$
$$H_U = \max\{1, \beta + 2(r-1)s\delta\}, \qquad H = \max\{H_P, H_U\}.$$

Here $\|\cdot\|_1$ denotes the coefficient ℓ_1 norm and an empty product equals one. With n_P, n_U, E as in the proof of Theorem 2.1, one may choose integer p_i, U_j in (5), with some $p_i \neq 0$, such that every coefficient has absolute value at most

$$\min\{H^{mE}, H_P^{n_P} H_U^{mn_U}\}. \tag{10}$$

This assertion bounds the operator of order at most r before any exact-order adjustment. For $m, t, s \ge 1$, the balanced pair in Proposition 2.2 gives

$$O\Big(m^3 t s^3 \left[h + \log(mts + 2)\right]\Big)$$

bits per coefficient, where $h = \log_2 \max\{1, \alpha, \beta, \delta\}$.

Proof. For polynomial rows use the sum of the coefficient ℓ_1 norms of their entries. Then $\|vA\|_1 \le \alpha\|v\|_1$, and recurrence (7) gives

$$\|v_{i+1}\|_1 \le (\alpha + 2it\delta)\|v_i\|_1.$$

Every telescoper column in the coefficient system has ℓ_1 norm at most H_P. A certificate column involves one row of B and one scalar derivative contribution; its norm is at most $\beta + 2(r-1)s\delta \le H_U$.

Choose independent columns and a nonsingular maximal-rank square submatrix. Setting one free coordinate to its determinant and solving for the pivot coordinates by the adjugate produces integer kernel vectors whose coordinates are maximal-rank minors. They span the rational kernel. Since that kernel has a nonzero telescoper projection, one such vector has a nonzero telescoper projection. Hadamard's inequality bounds a minor by H^{mE} and, by counting the two types of columns, by $H_P^{n_P} H_U^{mn_U}$. If the matrix has rank zero, use a unit vector in a telescoper coordinate instead.

For the balanced choice, $n_P = O(m^2 t s^2)$ and $mn_U = O(m^3 t s^3)$, while $\log H_P = O(ms[h + \log(mts + 2)])$ and $\log H_U = O(h + \log(ms + 2))$. Taking logarithms of the second bound in (10) proves the assertion. □

2.4 Relation to the original problem and prior work

Theorem 2.1 covers the full class of finite-rank *differential* D-finite presentations, rather than a selected function family. Its constants can be calculated directly from derivative multiplication matrices before telescoping. This supplies an explicit sufficient order–degree curve for the integration setting of original Problem 3.

The generalized Apagodu–Zeilberger theorem of Chen, Kauers, and Koutschan [2] already uses these multiplication matrices and bounds the dimension of the certificate-only kernel. Its ansatz has coefficients in $\mathbb{K}(x)$ and gives an order bound. The calculation here bounds the unknown coefficients polynomially in x, counts both degrees, and yields (6). Thus the formula is an explicit bidegree refinement of an established method. Independent verification of this derivation does not establish absence of an identical prior bound or a first-resolution claim. Since the original problem's surrounding discussion also includes summation, the result should not be described as resolving all shift, q-shift, or mixed variants.

3 Construction and worked identities

3.1 A finite algorithm

An effective implementation follows the proof directly.

1. Compute A, B, D from a derivative basis or an annihilating ideal, and read off m, t, s.
2. Choose $r \geq m(s+1)$ and $d = \lfloor c(r) \rfloor$, or choose the balanced pair of Proposition 2.2.
3. Generate $v_0, \ldots, v_r$ by (7) and expand the coordinate equations (8) in x and y.
4. Compute a basis of the homogeneous coefficient kernel. Select a vector with a nonzero telescoper projection, which the dimension argument guarantees exists.
5. Recover P, Q and verify $P - \partial_y Q$ by normal form modulo the original ideal. Raise the order by left multiplication only if an exact requested order is needed.

This gives an algorithm over an effectively computable field. The curve is measured after construction of the connection presentation; no bound on the cost of converting arbitrary input PDE generators is included. The proof also works with an annihilating ideal strictly smaller than the full annihilator, though its rank and degree bounds may then be less economical.

3.2 Rank two: an exponential times an Airy solution

Let $g''(y) = yg(y)$ and set

$$f = e^{xy} g(y), \qquad F = \begin{pmatrix} e^{xy} g(y) \\ e^{xy} g'(y) \end{pmatrix}.$$

The formal differential module has the compatible presentation

$$D = 1, \qquad A = yI_2, \qquad B = \begin{pmatrix} x & 1 \\ y & x \end{pmatrix}, \qquad m = 2,\ t = s = 1.$$

Since $F_2 = (\partial_y - x)f$, the certificate belongs to the required derivative module. Directly,

$$(\partial_x - x^2)f = \partial_y(F_2 - xf) = \partial_y((\partial_y - 2x)f). \tag{11}$$

For this presentation the uniform curve is $c(r) = (6r-2)/(r-3) = 6 + 16/(r-3)$, for $r \geq 4$. Some guaranteed degree bounds are

r	4	5	6	7	8	20
$\lfloor c(r) \rfloor$	22	14	11	10	9	6

Identity (11) is much smaller than these sufficient bounds. This illustrates why a valid general curve should not be confused with the optimal curve for a particular function.

3.3 A quadratic Airy pullback

Put $z = x + y^2$ and let $g''(z) = zg(z)$. For $F = (g(z), g'(z))^T$, one has

$$A = \begin{pmatrix} 0 & 1 \\ z & 0 \end{pmatrix}, \qquad B = 2yA, \qquad D = 1.$$

The parameters $m = 2, t = 1, s = 3$ are valid. Writing $f = g(z)$, the Airy equation gives $f_{xxx} = f + zf_x$, and therefore

$$(2\partial_x^3 - 2x\partial_x - 1)f = f + 2y^2 f_x = \partial_y(yf). \tag{12}$$

This identity has order three and degree one. The general theorem, which does not inspect the Airy equation's special cancellations, starts at $r = 8$ and gives $d = 106$ there. Equation (12) fits the theorem's ansatz at that order after zero padding.

3.4 A moving denominator in a Bessel pullback

Let g solve the Bessel equation $zg'' + g' + zg = 0$ and retain $z = x + y^2$. For $F = (g(z), g'(z))^T$ a polynomial common-denominator presentation is

$$D = z, \qquad A = \begin{pmatrix} 0 & z \\ -z & -1 \end{pmatrix}, \qquad B = 2yA.$$

Again $m = 2, t = 1, s = 3$. The certificate identity is

$$(2x\partial_x^3 + 3\partial_x^2 + 2x\partial_x + 1)f = \partial_y\left(\frac{y}{z}\partial_x f\right). \tag{13}$$

For verification, use $f_{xx} = -f - f_x/z$ and $f_{xxx} = -f_x + f/z + 2f_x/z^2$. Both sides then equal

$$-\frac{2y^2}{z}f + \left(\frac{1}{z} - \frac{4y^2}{z^2}\right)f_x.$$

Thus the rational certificate is valid as a differential-module identity, including its moving algebraic denominator. No assertion about evaluating an improper integral across a certificate pole follows automatically.

3.5 From identities to definite integrals

For fixed endpoints a, b, whenever differentiation under the integral and the endpoint evaluations are justified, $Pf = \partial_y(Qf)$ yields

$$P\int_a^b f(x, y)\, dy = (Qf)(x, b) - (Qf)(x, a).$$

A homogeneous equation for the integral requires those boundary terms to vanish or to be handled separately. The theorem is an algebraic creative-telescoping statement; it does not impose convergence or boundary conditions on arbitrary branches of its input functions.

4 Exact checks and reproducibility

The first embedded script constructs the entire coefficient system over $\mathbb{Q}$, computes its nullspace, selects a nonzero telescoper projection, and checks both the cleared polynomial identity and the uncleared rational connection identity. It separately computes the certificate-only kernel and checks the column norm bounds used in Proposition 2.4. All seven cases below passed. The actual kernel dimensions are computed over $\mathbb{Q}$; zero equation rows are counted in the stated matrix sizes.

Presentation	(m,t,s)	(r,d)	Matrix size	Pure kernel
Constant	$(1,0,0)$	$(1,0)$	1×3	1
Sine translate	$(2,0,0)$	$(2,0)$	2×5	0
Rank-three translate	$(3,1,0)$	$(3,9)$	39×76	0
Airy translate	$(2,1,1)$	$(4,22)$	270×323	0
Exponential times Airy	$(2,1,1)$	$(4,22)$	270×323	0
Bessel translate	$(2,1,1)$	$(4,22)$	270×323	0
Nilpotent module	$(2,0,0)$	$(2,0)$	2×5	1

The translate examples use $g(x+y)$ with the defining sine, Airy, or order-zero Bessel equation. The rank-three example uses $g(y+x^2/2)$ with $g^{(3)}=g'+g$. The nilpotent example is the formal cyclic module with $\partial_x F=0$ and $\partial_y F=\left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)F$; its rank is that of the supplied presentation, not necessarily of a particular scalar solution.

The second script verifies (12), (13), and the product-pullback identity $(x\partial_x+1)g(xy)=\partial_y(yg(xy))$ in their rank-two differential modules. All three residual vectors are exactly zero. These calculations use symbolic rational arithmetic, not numerical values of Airy or Bessel functions. Their role is to check examples and the implemented construction; the general theorem is proved by the dimension argument, not by these tests.

Single-source use. The TeX source contains its bibliography and two Python programs in `filecontents*` blocks. A normal compilation writes the programs to the compilation directory. With Python and SymPy installed, run

```
python3 verify_dfinite_curve.py
python3 verify_special_function_certificates.py
```

The programs emit exact-check results and save JSON records. Compiling the PDF does not execute Python or require shell escape. No separate bibliography database, figure file, or source fragment is needed.

5 Scope of the outcome

The established statement is an explicit sufficient curve, with a full proof, for all bivariate differential D-finite inputs in a finite-rank connection presentation. It extends the rank-one count to arbitrary rank and gives certificate and arithmetic bounds. It does not claim minimum order, optimal degree, basis invariance, or a curve for arbitrary shift and mixed systems.

The historical problem predates this manuscript, and the differential target has not been replaced by a specially chosen function family. However, a new underlying principle and absence of identical earlier solutions have not been established. Existing methods already supply the main mechanism and broader degree-control results. Thus this note does not meet the stronger standard of a certified new closure of a still-open named problem. It records the precise result that has been proved, with the attribution and limitations needed to assess it.

References

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